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Education

Huazhong University of Science and Technology

Wuhan, China

Ph.D. in Control Science & Engineering, School of Artificial Intelligence and Automation

- Advisor: Prof. Yan-Wu Wang

Nanyang Technological University

Singapore

Visiting Scholar/Student, School of Electrical and Electronic Engineering

Jan. 25, 2026 - Present

- Advisor: Prof. Hung Dinh Nguyen and Prof. Changyun Wen

Taiyuan University of Technology

Taiyuan, China

B.S. in Electrical Engineering, College of Electrical and Power Engineering

June 2020

- Advisor: Prof. Xiao-Ming Chang

Research Interests

Predefined-time Optimization and Control for Hybrid AC/DC Microgrid Clusters

- Predefined-Time Stability and Distributed Control Methods
- Multi-Objective Optimization for Power Flow Management
- Physics-Informed Neural Networks for Inertia Aggregation

Research Statement

With the increasing demands for electrical energy in different kinds of circumstances and the popularization of renewable energy resources, *microgrid*, a *micro-scale power system* consisting of distributed generators, bus and loads, is becoming more and more universal. The popularity of distributed generators and energy storage equipment makes application scenarios of microgrids more extensive.



Figure 1: An example of microgrid cluster.

To promote the efficient consumption of new energy and ensure the work safety of loads, it is necessary to build AC-DC mixing networked microgrids with high reliability. In the following, main achieved studies on *predefined-time optimization and control for hybrid AC/DC microgrid clusters* are listed.

I. Predefined-Time Economic Dispatch [JP1]

I.A Main Results

1. A new **smooth reconstruction penalty function** with continuous and piecewise linear differential is designed to deal with generation power constraints, which promotes to obtain a better suboptimal solution compared with the existing smooth penalty methods.

$$p_{\epsilon,i}(P_i) = \begin{cases} 0, & h_i(P_i) \leq 0 \\ \frac{k_\epsilon}{2} h_i^u(P_i), & 0 < h_i(P_i) \leq \epsilon_i, P_i > P_i^u \\ \frac{k_\epsilon}{2} h_i^l(P_i), & 0 < h_i(P_i) \leq \epsilon_i, P_i < P_i^l \\ h_i(P_i) - \frac{\epsilon_i - \Delta\epsilon_i^2}{2}, & h_i(P_i) > \epsilon_i \end{cases} \quad (1)$$

where $h_i(P_i) = (P_i^l - P_i)(P_i^u - P_i)$, $h_i^u(P_i) = (P_i^u - P_i)^2$ and $h_i^l(P_i) = (P_i^l - P_i)^2$. $\epsilon_i = h_i(P_i^u + \Delta\epsilon_i) = (1 + \tau)\tau P_i^{ul2}$ with $\Delta\epsilon_i = \tau P_i^{ul}$ and $P_i^{ul} = P_i^u - P_i^l$. τ is a positive scalar that regulates optimization error, and $k_\epsilon = \frac{P_i^{ul} + 2\Delta\epsilon_i}{\Delta\epsilon_i} = \frac{1+2\tau}{\tau}$.

2. A distributed predefined-time optimal economic dispatch strategy is presented by utilizing a time-based function. By employing the proposed strategy, the minimization of the generation cost with transmission loss under the power balance constraint and generation minimum/maximum constraints can be realized within a predefined settling time.

I.B Positive Comments

1. **Prof. Guan Xiaohong**, Xi'an Jiaotong University, China, Academician of the Chinese Academy of Sciences: "In [20], a time-based prespecified-time distributed approach was presented to figure out the optimal ED problem, which can simultaneously achieve the optimality of the solution and the fast convergence of the algorithm. ... a smooth reconstruction penalty function from [20] is introduced to cope with the inevitable equality/inequality constraints of participants in the microgrid."

2. **Prof. Zhi Tian**, *George Mason University, USA, IEEE Fellow*: “The economic dispatch problem for traditional thermal generations has been studied [4], where the generation costs of wind generation stations are time-varying due to their susceptibility to weather conditions.”

I.C Patent and Awards

1. **China Invention Patent**: A Distributed Optimal Economic Dispatch Method and System for Micro-grids, Publication Date: April 2, 2024, Patent Number: CN 117811079 A, Published.
2. 2024 Hubei Province **Excellent Science and Technology Paper**.
3. 20th National Conference on Complex Networks: **First Prize for Best Student Paper**, October 27, 2024, Nanjing, China.

II. Predefined-Time Cooperative Control [JP2-JP3 & CP1]

II.A Main Results

1. A distributed predefined-time controller (DPTC) is designed to achieve the control objectives of precise AC/DC bus voltage/frequency restoration and global power sharing among DGs in hybrid microgrid with unknown load power by **employing a class K^1 function and defining a unified error**. The convergence time of bus voltage and output power of converters can be adjusted by a predefined parameter.

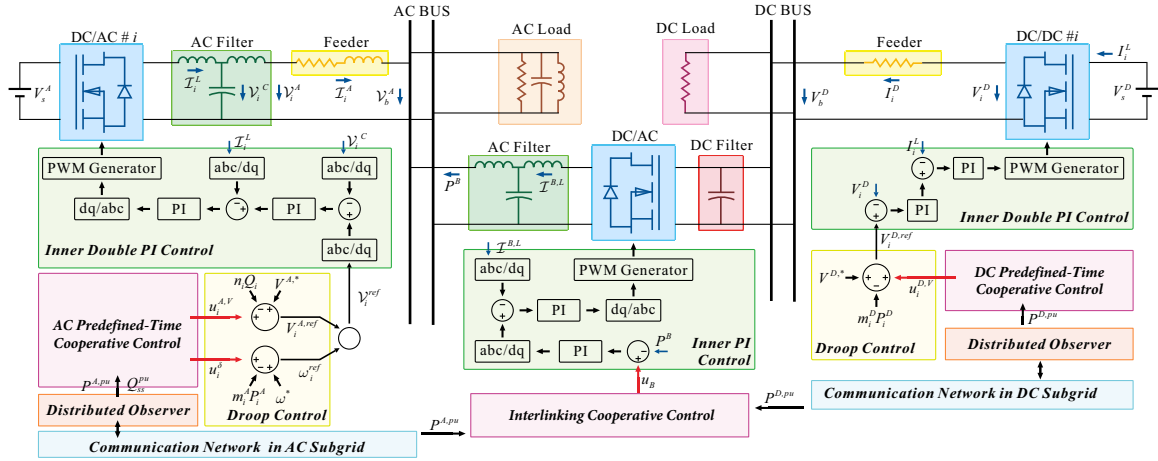


Figure 2: Control framework of hybrid microgrid.

2. A distributed predefined-time observer is proposed to rely less on direct load measurements and guarantee the controller implementation in emergencies when load information is unavailable. Comparative experiment indicates that the proposed controller has the potential for **convenient adjustment of convergence time and the advantage of small overshoot**.

II.B Positive Comments

1. **Prof. J. M. Guerrero**, *Aalborg University, Denmark, IEEE Fellow*: “These facts make the DC MG solution an attractive approach. In particular, distributed control is gradually becoming the dominant control mode in MGs because of its scalability, communication resource conservation, and resilience to single points of failure [10].”
2. **Prof. S. Bhattacharya**, *North Carolina State University, USA, IEEE Fellow*: “Several distributed control methods that neither rely on the prior estimates of cable resistances nor have the limitations of the undesired low-frequency ac ripple have been proposed [17], ... These methods typically utilize a low-bandwidth communication network to exchange information. The methods proposed in [17] communicated only load current to all the converters.”

II.C Patent and Applications

1. **Chinese Invention Patent**: A Secondary Control Method and Controller for Ensuring Stable Operation of a DC Microgrid, Grant Date: May 17, 2022, Patent Number: ZL202110837323.6.
2. **Huichuan Cup National Intelligent Automation Innovation Competition: National Third Prize**, August 2024, Suzhou, China.
3. **The 34th China Control and Decision Conference: Most Popular Academic Work Award**, August 2022, Hefei, China.

III. Power Flow Optimization [JP4-JP6 & CP2]

III.A Main Results

1. To characterize the trade-off between power sharing and transmission loss minimization in an interconnected AC microgrid, this paper designs a new **phase angle based multi-objective performance index**. A distributed practical predefined-time optimization algorithm is subsequently proposed to obtain the global optimal phase angle for minimizing the objectives-related performance index. Then, predefined-time observers are designed to promote the distributed implementation of the optimization algorithm. Under the optimal phase angle control strategy, the system's phase angles and output power of IBDGs can converge within a predefined settling time.

$$L(\theta) = 2(1 - \lambda_s)V_r^2 1_m^\top G_d(1_m - \cos\theta) + \lambda_s \kappa_s \mathcal{E}^\top(\theta)\mathcal{E}(\theta),$$

where $\mathcal{E}(\theta) = -V_r^2 \alpha_s^{e,-1}(\mathcal{D}B_d \sin\theta - G^E 1_n)$ with $\theta = \mathcal{D}^\top \delta$. $0 < \lambda_s < 1$ are adjustable coefficients regulating the weight of power sharing in the performance index. Additionally, κ_s is a positive constant representing the gain of class-normalized transformation from linear power sharing errors to quadratic form.

2. An **ϵ -exact penalty-based scalarization method** to solve the constrained multi-objective optimization problem of load sharing and transmission loss minimization within voltage safety constraints in a meshed DC microgrid. A distributed predefined-time optimization algorithm is designed and implemented by deploying consensus-based observers to obtain an optimal solution.

III.B Positive Comments

1. **Prof. F. Blaabjerg**, Aalborg University, Denmark, Fellow of the Danish Academy of Technical Sciences: "Based on a study in [43], a predefined-time-based control scheme of multi-bus DC MGs is proposed in order to control bus voltage, provide accurate current distribution, and reduce transmission losses. This control scheme is guaranteed to achieve control goals at a predefined time through a combined use of a distributed optimization and a predefined-time observer."

III.C Patent and Awards

1. **Chinese Invention Patent**: A Distributed Preset Time Optimization and Control Method for Multi-Bus DC Microgrids, Publication Date: November 17, 2023, Patent Number: CN 117081030 A.
2. **Advances in Engineering: Key Scientific Papers**.

IV. Predefined-Time Resilient Control [JP7]

IV.A Main Results

1. A **practical predefined-time resilient control strategy** is proposed for bus voltage regulation and current sharing among generators under DoS (denial-of-service) attacks and FDI (false data injection) attacks in not only to a single DC microgrid but also to networked DC microgrid cluster.
2. A **switching adaptive compensation algorithm** is developed to resist unbounded FDI attacks. The effects of attacks and the control parameters to regulation error and convergence time are derived theoretically for both SMG and NMG.

Journal Paper

Cited as JPx

1. **Yu Zhang**, Yan-Wu Wang, Jiang-Wen Xiao, and Xiao-Kang Liu. Distributed Predefined-Time Optimal Economic Dispatch for Microgrids. *Automatica* (Regular Paper), 2024, 169: 111870.
2. **Yu Zhang**, Yan-Wu Wang, Jiang-Wen Xiao, and Xiao-Kang Liu. Predefined-Time Secondary Control for DC Microgrid. *IEEE Transactions on Industrial Electronics*, 2022, 69(12): 13504-13513.
3. **Yu Zhang**, Yan-Wu Wang, Xiao-Kang Liu, Wu Yang, and Shu-Ming Liang. Distributed Predefined-Time Control for Hybrid AC/DC Microgrid. *IEEE Transactions on Industrial Electronics*, 2023, 70(8): 8324-8333.
4. **Yu Zhang**, Yan-Wu Wang, Xiao-Kang Liu, Guan-Nan Yu, and Yuehua Huang. Optimal phase angle control for interconnected AC microgrids. *Automatica* (Regular Paper), 2026, 183: 112654.
5. Yan-Wu Wang, **Yu Zhang**, Xiao-Kang Liu, and Xia Chen. Distributed Predefined-Time Optimization and Control for Multi-Bus DC Microgrid. *IEEE Transactions on Power Systems*, 2024, 39(4): 5769-5779.
6. **Yu Zhang**, Yan-Wu Wang, and Xiao-Kang Liu. Distributed Load-sharing and Loss Optimization within Voltage Safety Constraints for Meshed DC Microgrid. *IEEE/CAA Journal of Automatica Sinica*, DOI: 10.1109/JAS.2025.125345.
7. **Yu Zhang**, Yan-Wu Wang, Zhichun Yang, and Xiao-Kang Liu. From Single to Networked: Practical Predefined-Time Resilient Control of DC Microgrids under DoS and FDI Attacks. *IEEE Transactions on Industry Applications*, DOI: 10.1109/TIA.2025.3619026.

Conference Paper

Cited as CPx

1. **Yu Zhang**, Xiao-Kang Liu, Zheng Wu, Lan-Tao Xing. Distributed Predefined-Time Secondary Control for AC Microgrid. *49th Annual Conference of the IEEE Industrial Electronics Society*, Singapore, 2023.
2. **Yu Zhang**, Ke-Xin Ma, Ming-Yu Zhu, and Xiao-Kang Liu. Optimal Voltage Scheduling under Safety Constraints for Multi-bus DC Microgrid. *14th Asian Control Conference*, Dalian, China, 2024.

Research Project

1. **Hosted** Huazhong University of Science and Technology Innovation Research Institute Technology Innovation Fund Project “Research on Optimal Frequency Control Strategy of Microgrid Based on Distributed Energy Storage”, 2024.09-2025.09.
2. **Participated** National Natural Science Foundation General Project “Game Coordination and Application Research of Multi-scale Networked Systems under Incomplete Information”, 2022.01-2025.12: multi-scale game coordination in multi-source coordinated power supply;
3. **Participated** National Natural Science Foundation Key Project “Distributed Cooperative Control and Intelligent Decision-making of Regional Energy Internet”, 2023.01-2027.12: regional energy Internet flow optimization and collaborative control.

Honors & Awards

1. **National Scholarship** for Doctoral Students, 2025.
2. **National Scholarship** for Master’s Students, 2022.
3. 2024 Hubei Province **Excellent Science and Technology Paper**, 2025.
4. 20th National Conference on Complex Networks: **First Prize for Best Student Paper**, 2024.
5. Advances in Engineering: Key Scientific Papers, 2024.
6. Huichuan Cup National Intelligent Automation Innovation Competition: National Third Prize, 2024.